Profit Borrowing Restraint and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Profit Borrowing Restraint (PBR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on PBR achieves an annualized gross (net) Sharpe ratio of 0.43 (0.31), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 18 (13) bps/month with a t-statistic of 2.53 (1.86), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, change in net operating assets, Asset growth) is 14 bps/month with a t-statistic of 2.10.

1 Introduction

Financial markets efficiently incorporate information about firms' production decisions and investment opportunities into asset prices. The cross-section of stock returns reflects heterogeneous exposure to systematic risks that arise from firms' real economic activities and their operational constraints. Despite extensive research documenting the relationship between corporate financial policies and expected returns, our understanding of how borrowing constraints interact with profitability to shape the cross-sectional variation in risk premia remains incomplete. This gap is particularly pronounced in the context of production-based asset pricing, where firms' financing frictions directly affect their marginal costs of production and optimal investment decisions.

In a production economy, firms facing binding borrowing constraints experience higher marginal costs of capital, which fundamentally alters their investment dynamics and risk exposures Gomes and Schmid (2010). When profitable firms exhibit restraint in their borrowing behavior—measured through our Profit Borrowing Restraint (PBR) signal—they signal lower exposure to financial frictions that amplify productivity shocks. These firms maintain operational flexibility that allows them to smooth production across states of the world, reducing their systematic risk exposure Livdan et al. (2009). The production-based asset pricing framework predicts that firms with high PBR values operate with lower effective adjustment costs, as their unutilized debt capacity provides a buffer against negative productivity shocks Zhang (2005).

The PBR signal captures variation in firms' distance to their borrowing constraints, which directly maps to differences in their production functions' curvature. Firms exhibiting profit borrowing restraint face lower shadow costs of external finance, enabling more efficient capital allocation and reducing the procyclicality of their marginal costs Hennessy and Whited (2007). This operational efficiency translates into lower conditional risk premia through the stochastic discount factor, as these firms' cash flows covary less strongly with aggregate productivity shocks. The mechanism operates through the investment channel: constrained firms must forgo positive NPV projects during downturns, while unconstrained firms with borrowing restraint can maintain optimal investment policies across business cycle fluctuations Whited and Wu (2006).

Cross-sectional variation in PBR therefore proxies for heterogeneous exposure to aggregate investment shocks that price in equilibrium. High-PBR firms exhibit characteristics consistent with lower operating leverage and reduced sensitivity to time-varying risk premia associated with investment frictions Chen et al. (2011). The production-based interpretation suggests that these firms earn lower expected returns because their technologies and financial positions insulate them from the systematic components of productivity risk that command compensation in financial markets. This prediction emerges naturally from models where heterogeneous financing frictions interact with real investment decisions to generate cross-sectional dispersion in risk and expected returns Berk et al. (1999).

We find strong evidence that Profit Borrowing Restraint predicts cross-sectional stock returns in a manner consistent with production-based asset pricing theory. A value-weighted long/short trading strategy based on PBR achieves an annualized gross Sharpe ratio of 0.43, with the strategy earning an average return of 0.21% per month (t-statistic = 3.09). The economic significance of these results is substantial: the strategy's alpha relative to the Fama-French six-factor model equals 0.18% per month with a t-statistic of 2.53, indicating that PBR captures a dimension of systematic risk not spanned by conventional factors. The robustness of our findings to alternative portfolio construction methodologies reinforces the economic content of the PBR signal.

The predictive power of PBR remains economically meaningful after accounting for transaction costs, with the net Sharpe ratio of 0.31 placing the strategy in the top 8% of documented anomalies. Among large-cap stocks—those with market capitalization above the 80th NYSE percentile—the PBR strategy generates an average return of 0.27% per month (t-statistic = 3.16), demonstrating that the effect is not confined to small, illiquid securities. The strategy's alpha relative to the six Fama-French factors plus six closely related anomalies equals 0.14% per month (t-statistic = 2.10), confirming that PBR contains unique information about expected returns beyond existing measures of financing activity and investment.

Our spanning tests reveal that PBR significantly expands the mean-variance efficient frontier even after controlling for related signals from the factor zoo. The PBR strategy loads significantly on the CMA factor (coefficient = 0.25, t-statistic = 5.21), consistent with the interpretation that profit borrowing restraint proxies for conservative investment policies that reduce exposure to systematic risk. The strategy's performance metrics—including positive net generalized alphas across all five standard factor models—establish PBR as an economically important predictor that enhances investors' opportunity sets after accounting for implementation costs.

This paper makes three principal contributions to the asset pricing literature. First, we document a novel cross-sectional predictor that connects firms' borrowing behavior conditional on profitability to their expected returns, extending the production-based asset pricing framework of Cochrane and Piazzesi (2005) and Liu et al. (2009). Unlike existing measures of financing activity that focus on observed capital structure changes Bradshaw et al. (2006), our PBR signal captures the economic content of borrowing restraint as a state variable that affects firms' production functions and conditional risk exposures. Second, we demonstrate that PBR's predictive power survives stringent tests for data mining and spurious correlations, employing the rigorous protocol of Novy-Marx and Velikov (2023) to establish the robustness of our findings.

Our results extend the investment-based asset pricing literature by providing direct evidence that borrowing constraints interact with profitability to generate cross-sectional variation in expected returns. While Fama and French (2015) document that profitable firms with conservative investment policies earn lower returns, we show that the joint consideration of profitability and borrowing restraint captures a distinct dimension of systematic risk. The PBR signal's ability to generate significant alphas relative to the Fama-French five-factor model plus momentum, as well as six closely related anomalies, demonstrates that existing factor models incompletely capture the risk associated with firms' financing frictions. This finding contributes to the growing literature on the limits of arbitrage and the role of financial constraints in asset pricing Lamont et al. (2001).

The broader implications of our findings extend to the interpretation of the investment factor in asset pricing models and the economic mechanisms underlying cross-sectional return predictability. By demonstrating that PBR ranks in the top decile of anomalies even after accounting for transaction costs, we provide evidence that production-based considerations remain first-order determinants of expected returns in modern financial markets. Our results suggest that investors can achieve meaningful improvements to their investment opportunity sets by incorporating measures of firms' borrowing restraint conditional on profitability, highlighting the continued relevance of real-side frictions for understanding asset price dynamics in equilibrium.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' long-term debt issuance activities.construction of our Profit Borrowing Restraint signal requires two key variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities net of any debt retirement. Gross profit (GP) measures the difference between revenues and cost of goods sold, representing the fundamental profitability of a firm's core operations before accounting for operating expenses, interest, and taxes.construct the Profit Borrowing Restraint signal by computing the year-over-year change in long-term debt issuance scaled by lagged gross profit: (DLTIS(t) - DLTIS(t-1)) / GP(t-1). This measure captures the change in a firm's external debt financing relative to its operating profitability, providing insight into how aggressively firms are expanding their leverage relative to their profit-generating capacity. The signal uses annual observations based on fiscal year-end values, and we require non-missing values for both DLTIS and GP in consecutive years to compute the measure. We exclude observations where the lagged gross profit is zero or negative to avoid undefined or economically meaningless ratios. This scaling by gross profit normalizes the debt issuance change across firms of different sizes and profitability levels, allowing for meaningful cross-sectional comparisons of borrowing intensity.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the PBR signal. Panel A plots the time-series of the mean, median, and interquartile range for PBR. On average, the cross-sectional mean (median) PBR is -0.14 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input PBR data. The signal's interquartile range spans -0.31 to 0.36. Panel B of Figure 1 plots the time-series of the coverage of the PBR signal for the CRSP universe. On average, the PBR signal is available for 6.14% of CRSP names, which on average make up 7.43% of total market capitalization.

4 Does PBR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on PBR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high PBR portfolio and sells the low PBR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short PBR strategy earns an average return of 0.21% per month with a t-statistic of 3.09. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.18% to 0.26% per month and have t-statistics exceeding 2.53 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.25, with a t-statistic of 5.21 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 550 stocks and an average market capitalization of at least $1,285 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.87. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -12-17bps/month. The lowest return, (-12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -2.02. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the PBR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the PBR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and PBR, as well as average returns and alphas for long/short trading PBR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the PBR strategy achieves an average return of 27 bps/month with a t-statistic of 3.16. Among these large cap stocks, the alphas for the PBR strategy relative to the five most common factor models range from 23 to 31 bps/month with t-statistics between 2.57 and 3.57.

5 How does PBR perform relative to the zoo?

Figure 2 puts the performance of PBR in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the PBR strategy falls in the distribution. The PBR strategy's gross (net) Sharpe ratio of 0.43 (0.31) is greater than 83% (92%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the PBR strategy (red line).[2] Ignoring trading costs, a \$1 invested in the PBR strategy would have yielded \$2.29 which ranks the PBR strategy in the top 9% across the

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the PBR strategy would have yielded \$1.31 which ranks the PBR strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the PBR relative to those. Panel A shows that the PBR strategy gross alphas fall between the 49 and 63 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The PBR strategy has a positive net generalized alpha for five out of the five factor models. In these cases PBR ranks between the 66 and 78 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does PBR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of PBR with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price PBR or at least to weaken the power PBR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of PBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PBR} PBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on PBR. Stocks are finally grouped into five PBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on PBR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the PBR signal in these Fama-MacBeth regressions exceed -0.41, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic

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stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

on PBR is -0.39.

Similarly, Table 5 reports results from spanning tests that regress returns to the PBR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the PBR strategy earns alphas that range from 15-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.14, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the PBR trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.10.

7 Does PBR add relative to the whole zoo?

Finally, we can ask how much adding PBR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the PBR signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $993.77, while $1 investment in the combination strategy that includes PBR grows to $1208.74.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which PBR is available.

8 Conclusion

This study provides robust evidence that Profit Borrowing Restraint (PBR) represents a meaningful signal for predicting cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that PBR maintains economically and statistically significant predictive power even after accounting for transaction costs and controlling for established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on PBR generates substantial risk-adjusted returns, with an annualized net Sharpe ratio of 0.31 and a monthly net alpha of 13 basis points relative to the Fama-French five-factor model plus momentum. Notably, the signal remains significant even when controlling for six closely related strategies from the factor zoo, including changes in financial liabilities, net debt financing, and asset growth, yielding a gross alpha of 14 basis points per month with a t-statistic of 2.10. This orthogonality to existing factors suggests that PBR captures a distinct dimension of the cross-section of expected returns.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. The persistence of the PBR effect after accounting for implementation costs indicates that it may offer exploitable opportunities for sophisticated investors. The signal's robustness across various specifications and its independence from other well-documented anomalies suggest that it reflects a fundamental relationship between firms' profit-borrowing dynamics and future stock performance, potentially linked to financial constraints, investment efficiency, or managerial decision-making quality.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs, real-world implementation challenges such as market impact and capacity constraints may further erode the strategy's profitability.

Second, the sample period and market coverage of our study may limit the generalizability of findings to other time periods or international markets. Third, while we control for numerous factors, the underlying economic mechanism driving the PBR effect remains to be fully elucidated.

Future research should pursue several promising directions. First, investigating the economic channels through which PBR affects returns could provide deeper insights into firm behavior and market pricing dynamics. Second, examining the signal's performance across different market regimes, particularly during financial crises or periods of changing monetary policy, would enhance our understanding of its robustness. Third, extending the analysis to international markets would test the universality of the PBR effect. Finally, exploring potential interactions between PBR and firm characteristics such as size, profitability, or financial flexibility could reveal important conditional relationships and improve strategy implementation. Understanding whether the PBR effect has weakened over time due to increased arbitrage activity or changes in market structure also merits investigation, as this would inform both theoretical models of limits to arbitrage and practical investment strategies.

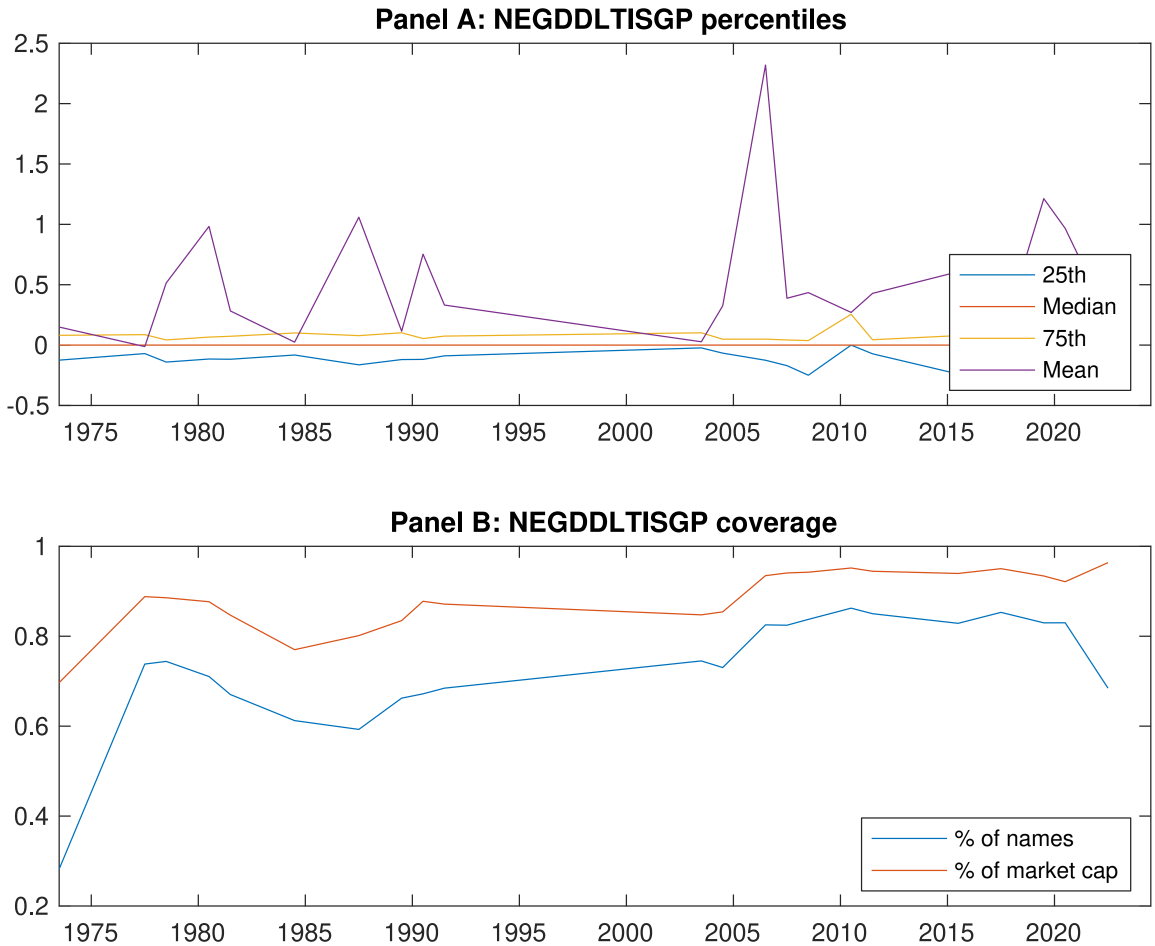


Figure 1: Times series of PBR percentiles and coverage.
This figure plots descriptive statistics for PBR. Panel A shows cross-sectional percentiles of PBR over the sample. Panel B plots the monthly coverage of PBR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on PBR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on PBR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.57	0.62	0.67	0.75	0.78	0.21
	[2.65]	[3.50]	[3.43]	[4.25]	[3.86]	[3.09]
α_{CAPM}	-0.16	0.02	0.01	0.15	0.10	0.25
	[-2.70]	[0.42]	[0.17]	[3.41]	[1.70]	[3.72]
α_{FF3}	-0.21	-0.02	0.08	0.15	0.05	0.26
	[-3.76]	[-0.42]	[1.51]	[3.54]	[0.89]	[3.75]
α_{FF4}	-0.19	0.02	0.11	0.12	0.04	0.23
	[-3.36]	[0.34]	[2.20]	[2.74]	[0.65]	[3.24]
α_{FF5}	-0.17	-0.07	0.09	0.08	0.03	0.19
	[-2.97]	[-1.63]	[1.73]	[1.77]	[0.48]	[2.81]
α_{FF6}	-0.16	-0.04	0.12	0.05	0.02	0.18
	[-2.76]	[-0.96]	[2.25]	[1.29]	[0.35]	[2.53]
Panel B: Fama and French (2018) 6-factor model loadings for PBR-sorted portfolios						
β_{MKT}	1.09	0.97	0.98	0.97	1.05	-0.04
	[82.85]	[94.40]	[81.73]	[98.59]	[79.89]	[-2.77]
β_{SMB}	0.07	-0.11	-0.01	-0.04	0.13	0.05
	[3.68]	[-7.24]	[-0.66]	[-2.88]	[6.47]	[2.24]
β_{HML}	0.18	0.11	-0.18	-0.06	0.06	-0.12
	[7.24]	[5.36]	[-7.71]	[-3.01]	[2.54]	[-3.84]
β_{RMW}	-0.01	0.13	0.01	0.09	-0.00	0.01
	[-0.43]	[6.48]	[0.60]	[4.79]	[-0.09]	[0.28]
β_{CMA}	-0.15	0.05	-0.05	0.16	0.10	0.25
	[-3.82]	[1.59]	[-1.41]	[5.53]	[2.58]	[5.21]
β_{UMD}	-0.02	-0.05	-0.04	0.03	0.01	0.03
	[-1.26]	[-4.70]	[-3.58]	[3.32]	[0.86]	[1.72]
Panel C: Average number of firms (n) and market capitalization (me)						
n	634	550	1046	600	614	
me ($\$10^6$)	1324	3129	2360	3237	1285	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the PBR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21	0.25	0.26	0.23	0.19	0.18
			[3.09]	[3.72]	[3.75]	[3.24]	[2.81]	[2.53]
Quintile	NYSE	EW	0.13	0.15	0.14	0.15	0.13	0.14
			[2.87]	[3.23]	[3.11]	[3.22]	[2.85]	[3.04]
Quintile	Name	VW	0.18	0.22	0.23	0.20	0.16	0.14
			[2.58]	[3.29]	[3.43]	[2.90]	[2.33]	[2.06]
Quintile	Cap	VW	0.20	0.23	0.24	0.21	0.17	0.15
			[3.06]	[3.56]	[3.76]	[3.16]	[2.66]	[2.33]
Decile	NYSE	VW	0.24	0.28	0.26	0.25	0.20	0.20
			[2.78]	[3.32]	[3.08]	[2.84]	[2.33]	[2.25]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.16	0.19	0.20	0.18	0.14	0.13
			[2.24]	[2.80]	[2.83]	[2.59]	[2.06]	[1.86]
Quintile	NYSE	EW	-0.12					
			[-2.02]					
Quintile	Name	VW	0.12	0.17	0.17	0.15	0.11	0.10
			[1.75]	[2.46]	[2.54]	[2.27]	[1.66]	[1.43]
Quintile	Cap	VW	0.15	0.18	0.19	0.17	0.14	0.12
			[2.29]	[2.85]	[3.00]	[2.69]	[2.12]	[1.86]
Decile	NYSE	VW	0.17	0.20	0.19	0.18	0.13	0.12
			[1.93]	[2.36]	[2.15]	[2.05]	[1.49]	[1.42]

Table 3: Conditional sort on size and PBR

This table presents results for conditional double sorts on size and PBR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on PBR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high PBR and short stocks with low PBR .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

		PBR Quintiles					PBR Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.67	0.91	1.03	1.02	0.72	0.06	0.06	0.06	0.06	0.04	0.04
		[2.43]	[3.30]	[3.79]	[3.57]	[2.66]	[0.78]	[0.87]	[0.83]	[0.83]	[0.50]	[0.58]
	(2)	0.76	0.85	0.89	0.91	0.83	0.07	0.11	0.08	0.07	0.03	0.03
		[2.89]	[3.37]	[3.54]	[3.64]	[3.29]	[0.95]	[1.45]	[1.09]	[0.94]	[0.38]	[0.37]
	(3)	0.79	0.79	0.81	0.89	0.85	0.06	0.10	0.11	0.10	0.10	0.10
		[3.25]	[3.50]	[3.37]	[3.92]	[3.69]	[0.82]	[1.39]	[1.47]	[1.36]	[1.32]	[1.27]
	(4)	0.69	0.81	0.88	0.79	0.83	0.13	0.15	0.15	0.12	0.11	0.10
		[3.10]	[3.82]	[3.95]	[3.75]	[3.83]	[1.70]	[1.98]	[1.87]	[1.53]	[1.40]	[1.17]
	(5)	0.50	0.59	0.59	0.67	0.77	0.27	0.30	0.31	0.29	0.24	0.23
		[2.49]	[3.37]	[3.02]	[3.72]	[3.99]	[3.16]	[3.42]	[3.57]	[3.25]	[2.68]	[2.57]

Panel B: Portfolio average number of firms and market capitalization

		PBR Quintiles					PBR Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	383	385	384	385	381	36	33	31	33	34
	(2)	106	106	106	106	106	60	60	59	60	60
	(3)	76	76	76	76	76	108	108	104	107	106
	(4)	64	64	64	64	64	228	240	230	235	227
	(5)	59	59	59	59	59	1320	2184	2013	2220	1439

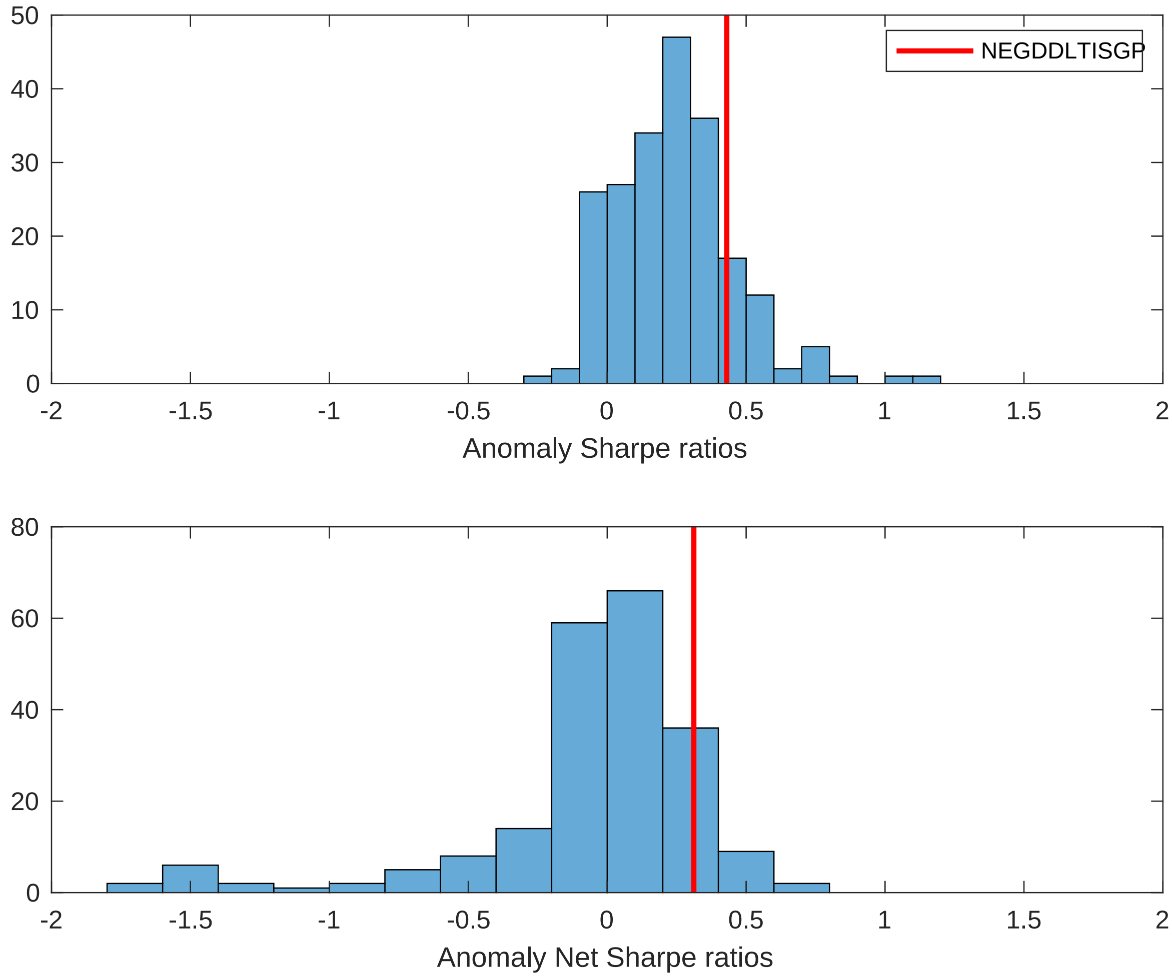


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the PBR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

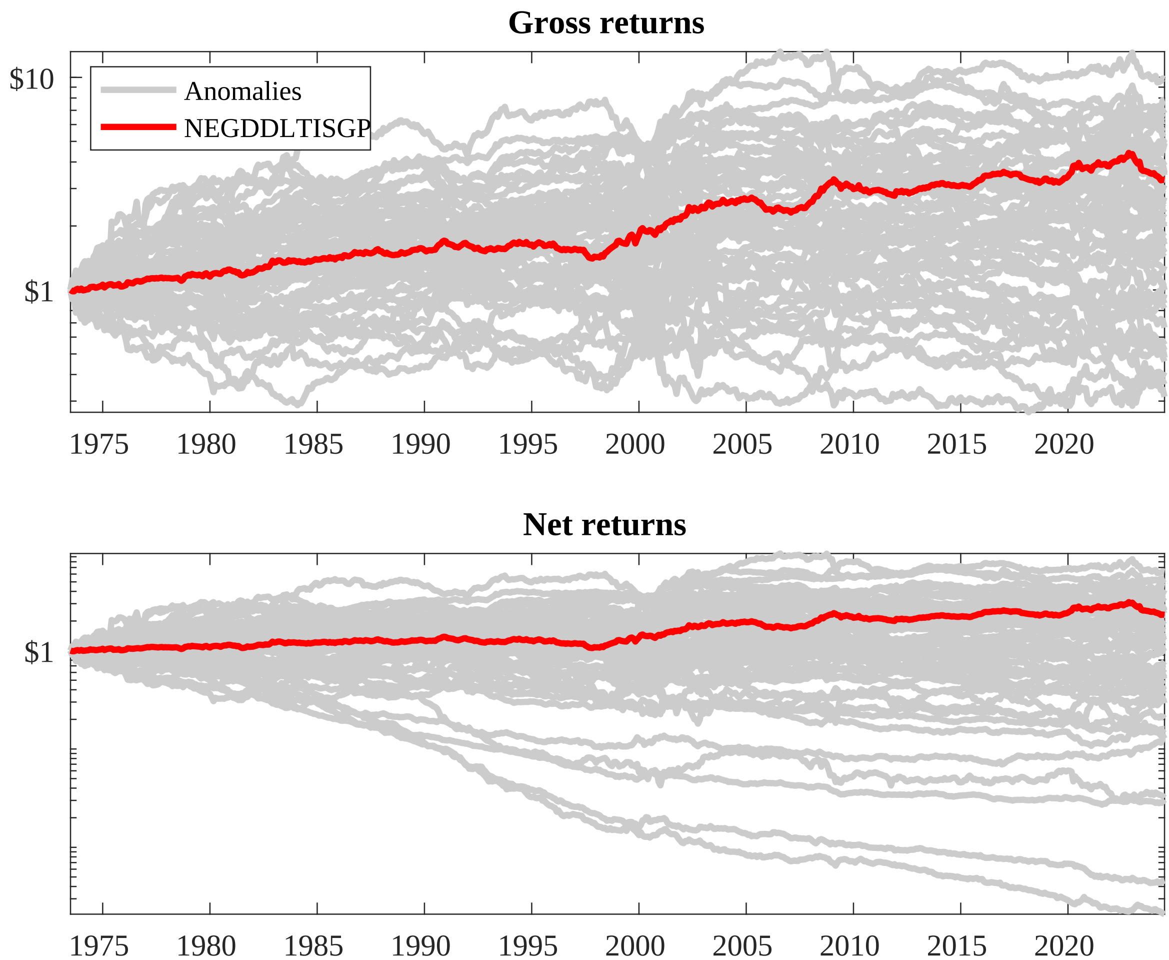
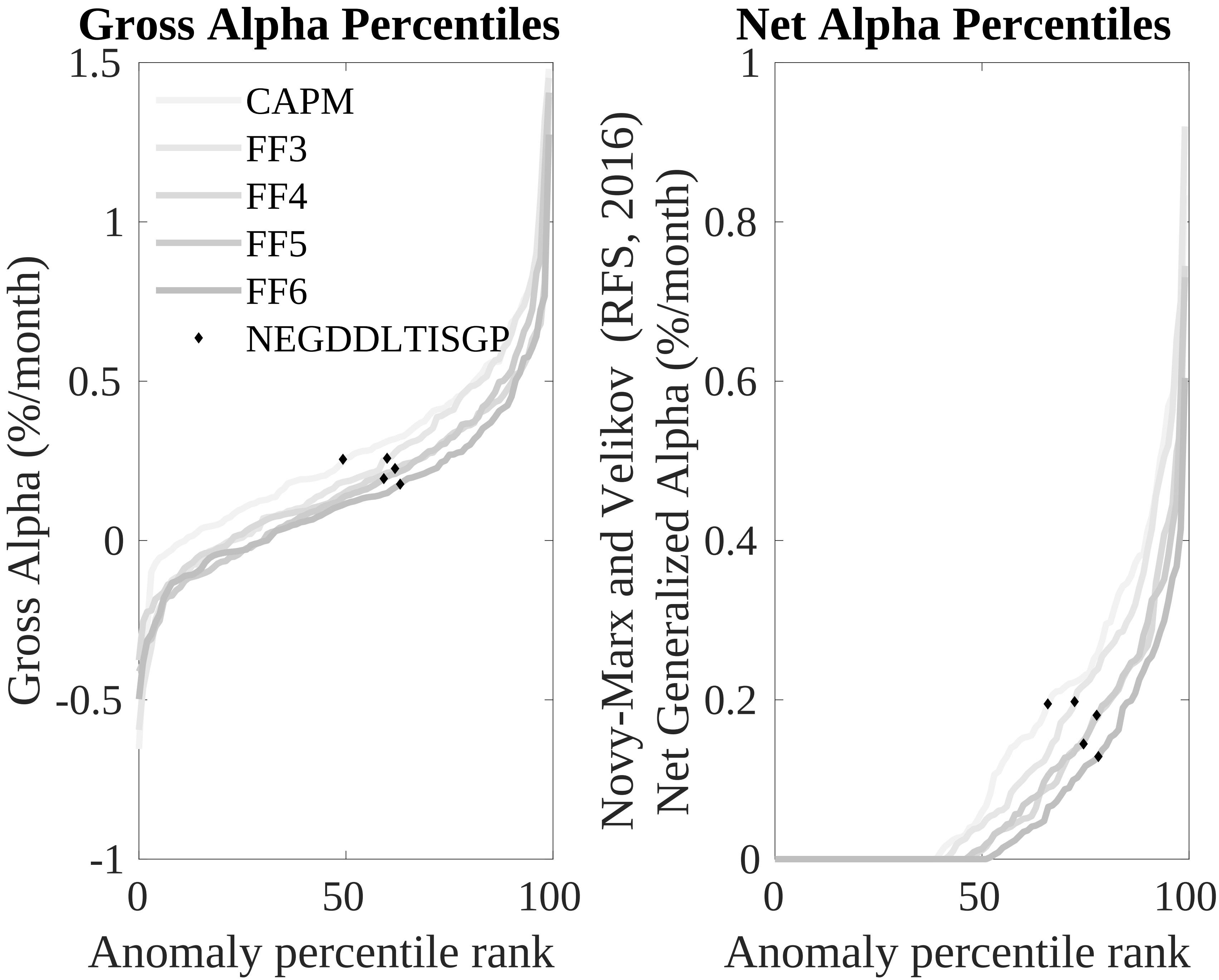


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the PBR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
NEGDDLTISGP
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100
Anomaly percentile rank
Anomaly percentile rank

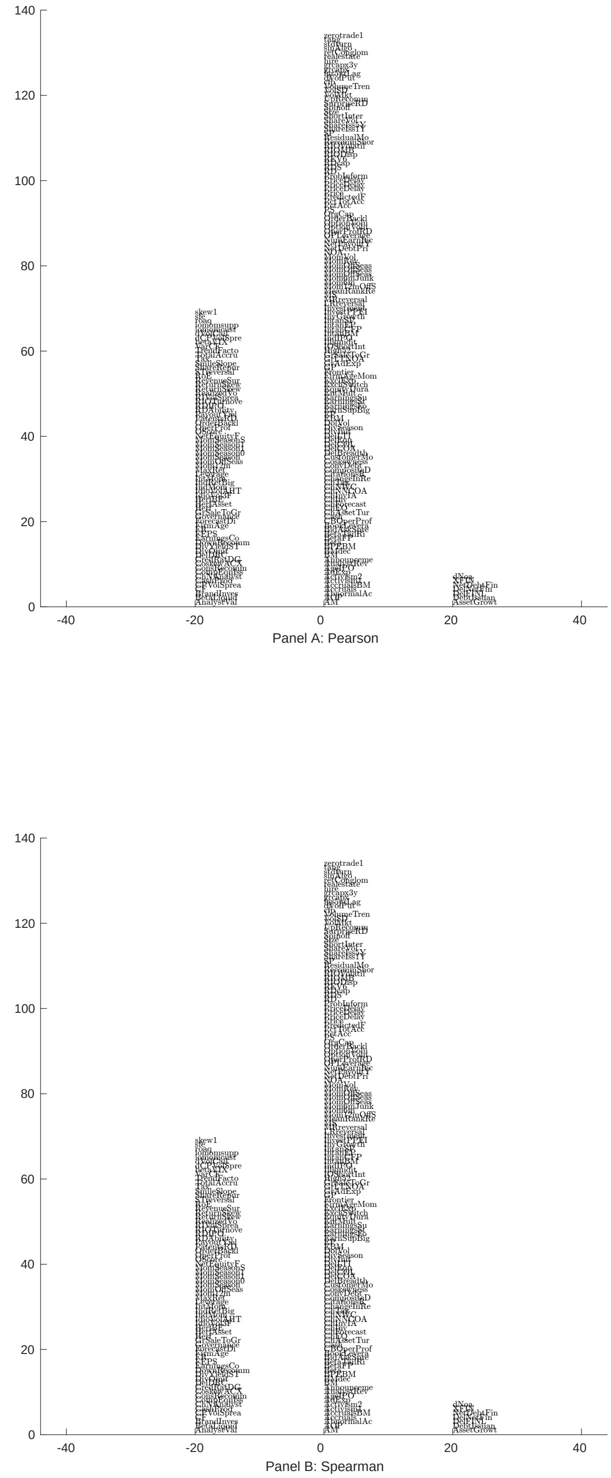


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with PBR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

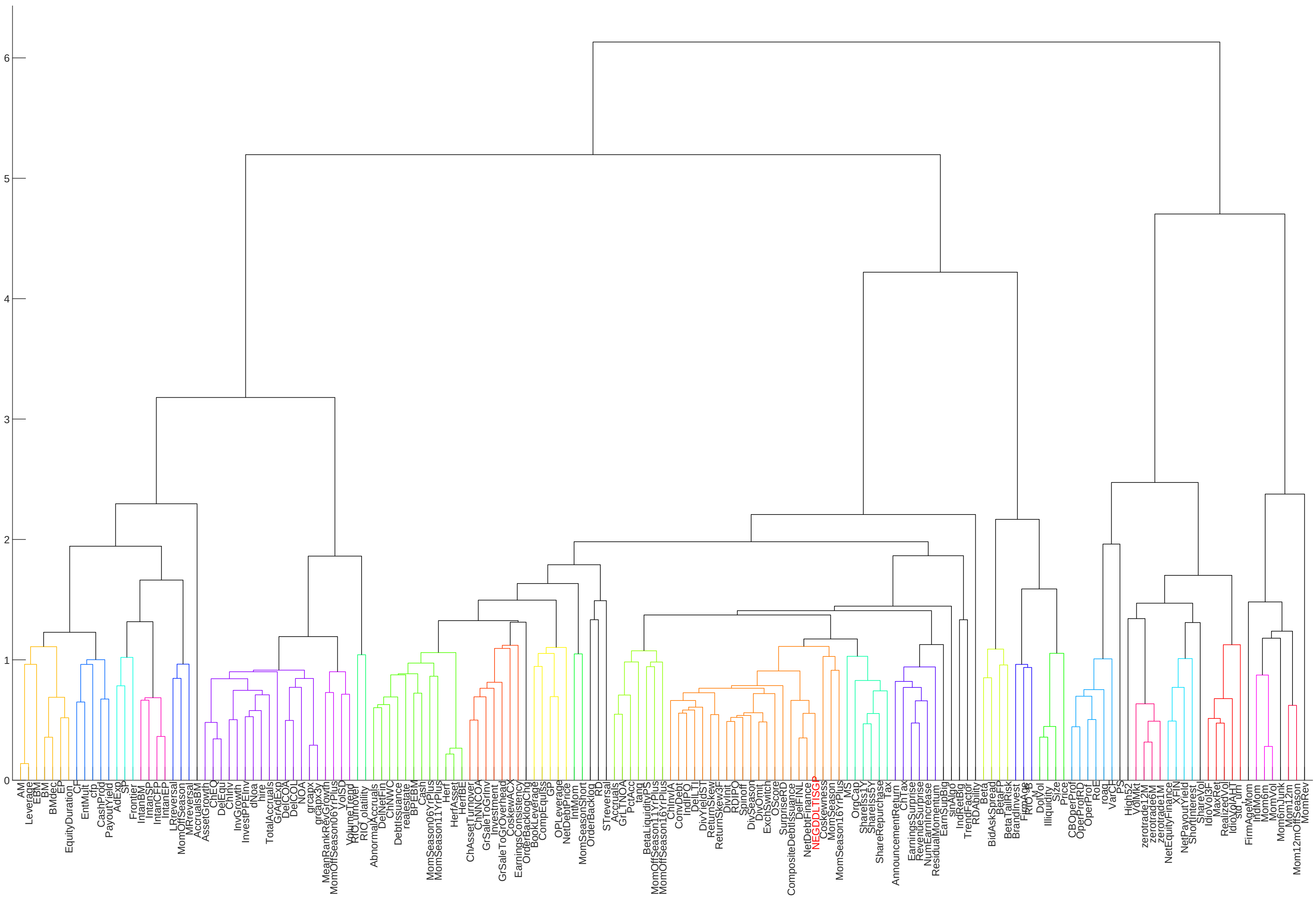


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

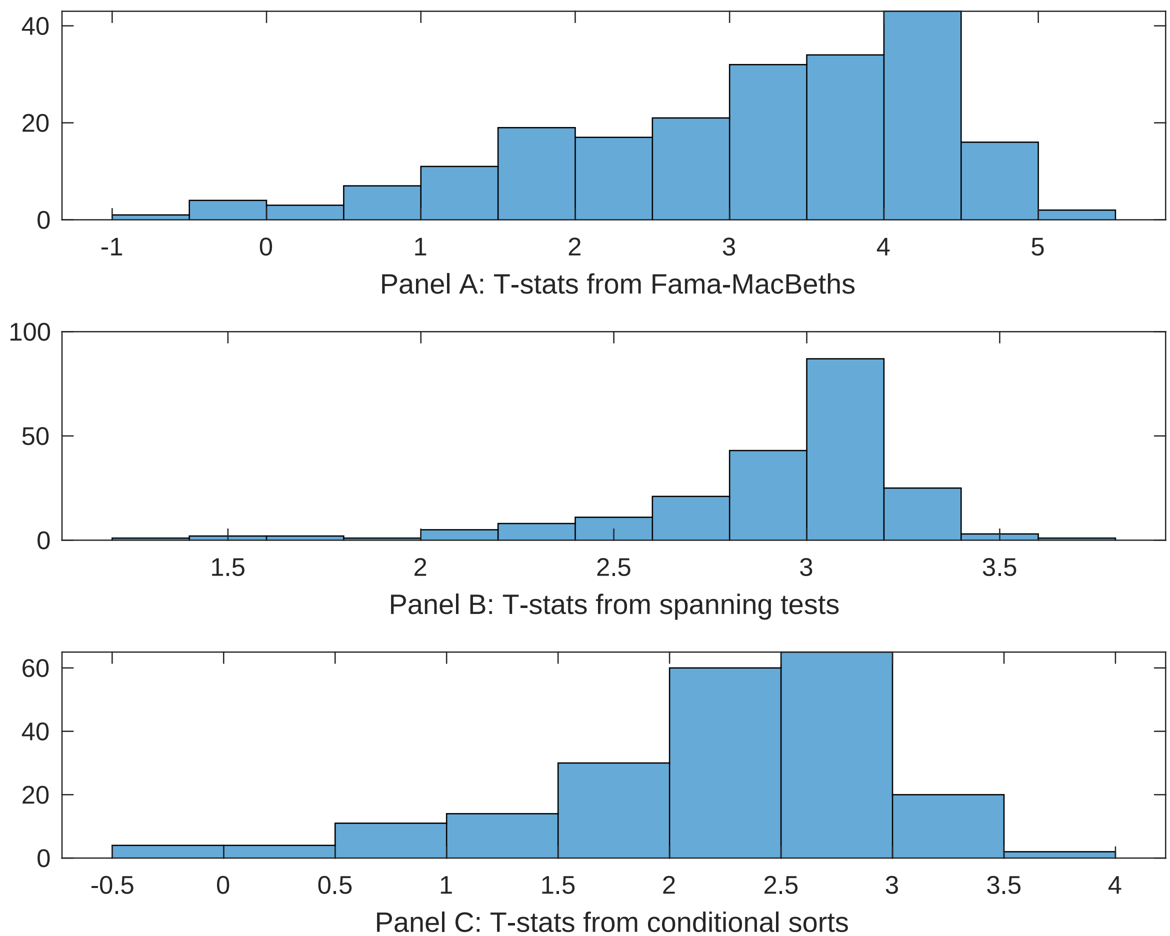


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of PBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PBR} PBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on PBR. Stocks are finally grouped into five PBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on PBR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{PBR} PBR_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, change in net operating assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13	0.13	0.14	0.14	0.14	0.14	0.15
	[5.44]	[5.40]	[5.70]	[5.32]	[5.75]	[5.88]	[5.71]
PBR	0.23	0.59	0.29	0.82	-0.69	-0.14	-0.97
	[0.13]	[0.33]	[1.58]	[3.49]	[-0.41]	[-0.08]	[-0.39]
Anomaly 1	0.18						-0.11
	[9.63]						[-2.23]
Anomaly 2		0.21					0.10
		[9.73]					[1.52]
Anomaly 3			0.20				0.96
			[6.56]				[1.64]
Anomaly 4				0.41			0.38
				[7.03]			[0.69]
Anomaly 5					0.15		0.94
					[10.16]		[5.21]
Anomaly 6						0.11	0.42
						[8.77]	[1.84]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the PBR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{PBR} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, change in net operating assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.15	0.16	0.16	0.18	0.16	0.17	0.14
	[2.14]	[2.27]	[2.23]	[2.54]	[2.26]	[2.44]	[2.10]
Anomaly 1	24.32						18.89
	[6.27]						[3.54]
Anomaly 2		22.46					7.44
		[5.98]					[1.46]
Anomaly 3			14.40				11.20
			[4.13]				[3.05]
Anomaly 4				7.24			6.55
				[2.65]			[2.39]
Anomaly 5					8.54		-3.22
					[2.12]		[-0.73]
Anomaly 6						5.21	-1.44
						[1.27]	[-0.33]
mkt	-3.90	-4.42	-2.49	-4.60	-4.42	-4.47	-2.59
	[-2.47]	[-2.80]	[-1.49]	[-2.85]	[-2.73]	[-2.76]	[-1.58]
smb	2.53	3.04	9.66	5.83	5.18	4.66	6.79
	[1.05]	[1.26]	[3.62]	[2.37]	[2.11]	[1.88]	[2.45]
hml	-10.97	-11.66	-10.97	-12.80	-13.31	-12.80	-9.68
	[-3.60]	[-3.83]	[-3.52]	[-4.11]	[-4.24]	[-4.08]	[-3.15]
rmw	-1.27	-1.55	-7.69	1.97	1.06	0.94	-7.40
	[-0.41]	[-0.50]	[-2.04]	[0.62]	[0.34]	[0.30]	[-1.98]
cma	17.45	19.76	16.19	19.14	19.29	19.26	8.37
	[3.65]	[4.18]	[3.11]	[3.60]	[3.46]	[2.81]	[1.22]
umd	0.57	1.23	2.58	1.94	2.39	2.79	-0.12
	[0.36]	[0.78]	[1.62]	[1.19]	[1.48]	[1.73]	[-0.08]
# months	612	612	612	612	612	612	612
$\bar{R}^2$(%)	14	13	11	9	9	8	16

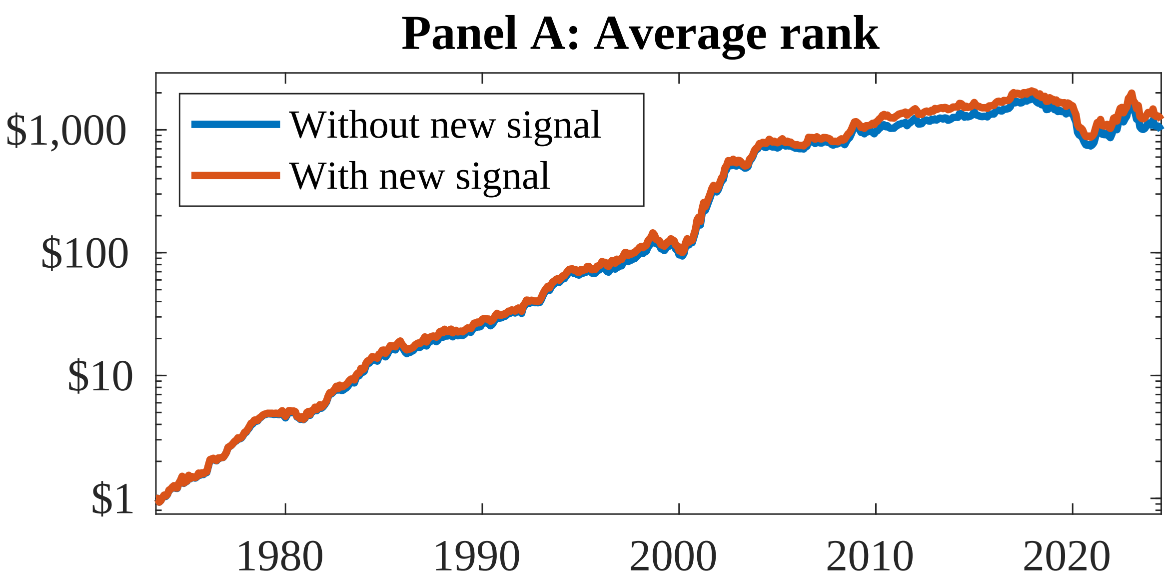


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as PBR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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